



MINING DEALER BEST PRACTICE

Procedure for Testing the Damper of the C-175 Engines

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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December 2016
4.03-161215-01

1.0 Introduction

The C175 Engine damper represented a challenge to our Component Rebuild Center in regards to the complexity to test this component and ensure the adequate condition before reusing it. The challenge to test the damper lies in the characteristics of the design and operating principles applied to the C175 damper.

The damping effect of this component is based on a combination of springs and hydraulics, which in this case use engine oil. In addition to the springs that provide the primary source of damping and control of the crankshaft vibration, the engine oil provides extra damping effect and also cools down the damper while in operation.

Our challenge was related to the testing of the hydraulic part of the damper, being able to ensure that this component doesn't leak and operates properly when installed in a rebuilt engine. The consequences of installing a faulty damper in a rebuilt engine and then into a 797 OHT are significant, and are related to Redo, MTTR, MTBF, MTBS, Availability and M&R cost of this type of equipment.

This Best Practice proposes a procedure we developed to test the hydraulic part of this damper, and ensure the appropriate condition of it before reusing it in a rebuilt engine.

2.0 Best Practice Description

The procedure for testing the C175 Engine Damper described in this Best Practice considers the application of pressurized air to the hydraulic chamber of the damper to detect leaks and evaluate it's sealing capability.



C175 damper in the process of being tested.

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In order to be able to apply this testing procedure we used a series of common tools and parts, and fabricated a special adapter that we will describe in this BP.

This is the list of tools and attachments that needs to be considered to implement this procedure.

Tools and Elements

Item #	Quantity	Elements Name	Parts Number
• 1	1	Adapter	Manufactured locally
• 2	1	Adapter Seal	176-1378
• 3	1	Air tightness tools	
• 4	1	Coupling male ¼" NPT	
• 5	4	Nuts (5/8-11-THD)	4K-0367
• 6	8	Washer (17.5X44.4X3.5-MM THK)	4B-5275
• 7	4	Bolts (5/8-11X10.5-IN)	5P-4626
• 8	1	Link Bracket ¾"	439-3939
• 9	1	Bolts ¾" (3/4-10X3-IN)	9X-8887
• 10	1	Nut ¾" (3/4-10-THD)	2J-3506
• 11	2	Washer (20X36.5X3.5-MM THK)	5P-8248
• 12	1	holder for lifting	

The photos below illustrate some of these tools that are needed.



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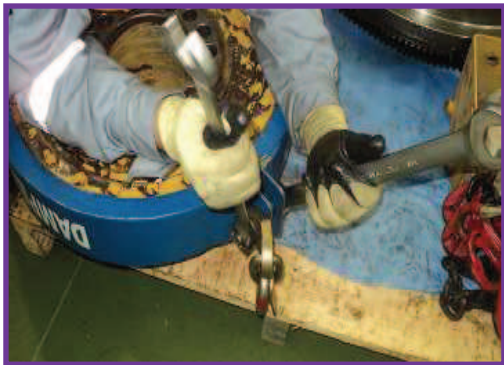
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Note: The adapter marked as “1” is shown in 3 different positions. It is only one adapter.

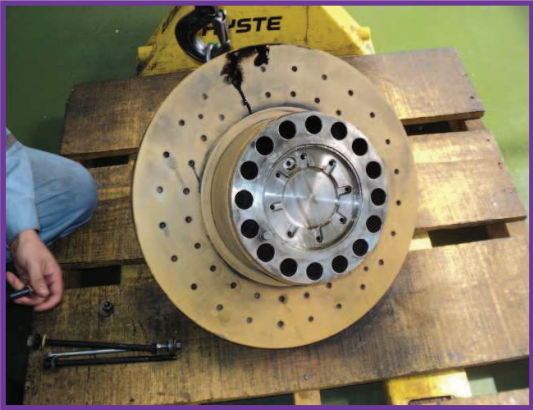
Testing Procedure

2.1 Install the holder (12) for lifting and maneuvering the damper safely.

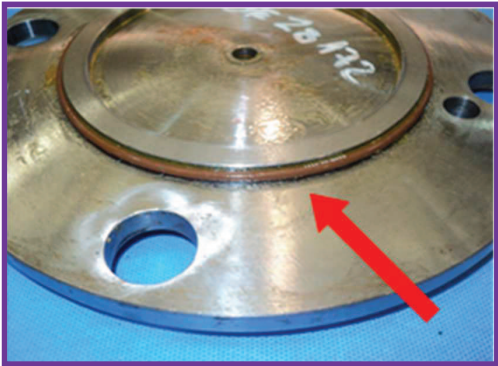


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2.2 Position the damper so that the technicians can work safely and comfortably.



2.3 Install seal retainer (2) to the adapter (1).



2.4 Install the adapter (1) on the damper with the four bolts (7) with washers (6) and nuts (5). Tighten the bolts to 45 foot-pound torque (ft-lb).



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2.5 Install 1/4 coupling (4), using Teflon tape to avoid possible leaks.

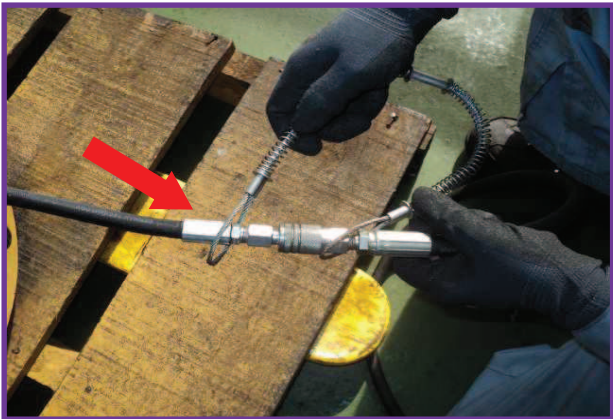
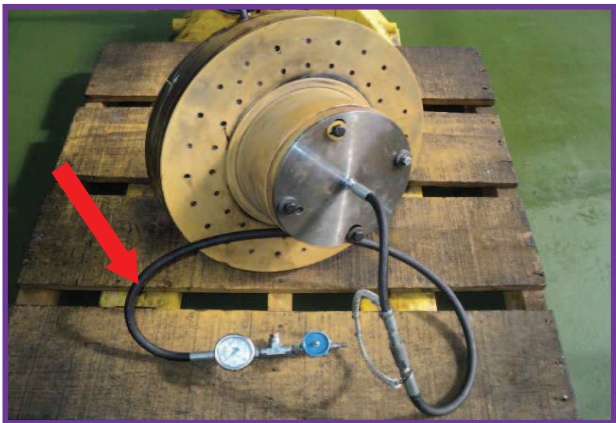


2.6 Set the pressure of the supply line to 50 PSI. This is the maximum pressure established for this test.



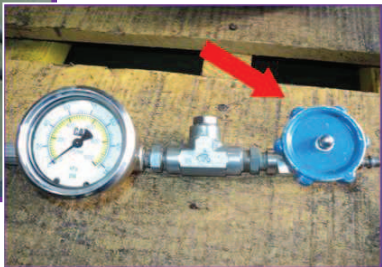
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2.7 Install the air tightness tool (3) to the adapter (1) and connect it to the to the air supply line.



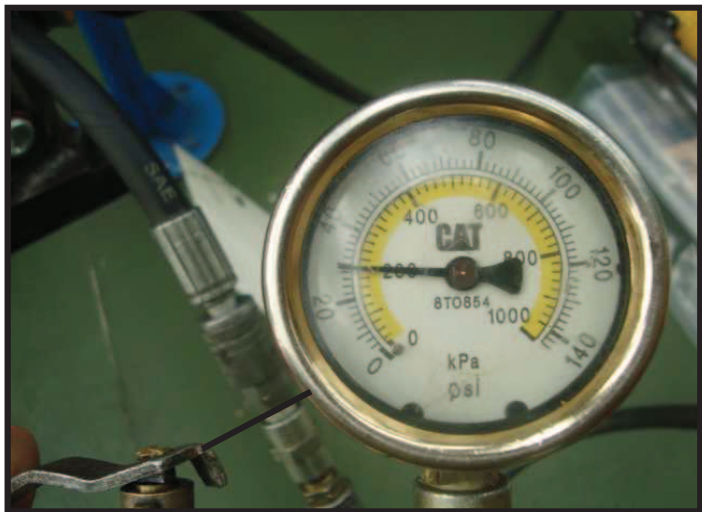
Air tightness tool installed to test the damper. Notice the use of whip-check to secure the hoses and avoid injuries.

2.8 Regulate the air pressure to 40 to 50 psi to perform the leakage test. Block the air supply and maintain and observe the pressure in the damper for 5 minutes.



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2.9 If after 5 minutes the pressure doesn't drop, the damper is considered ok and can be reused.



Consider the use of a mix of water and soap to identify possible air leaks

Warning: At the completion of the test, release the air pressure to return the damper a “zero energy” condition.

3.0 Implementation Steps

- 3.1 Acquisition of the items / tools recommended**
- 3.2 Fabricate the plate adapter**
- 3.3 Practical testing**
- 3.4 Document the Standard Operating Procedure (SOP)**
- 3.5 Train the personnel**
- 3.6 Release the procedure to operation and validate benefits**

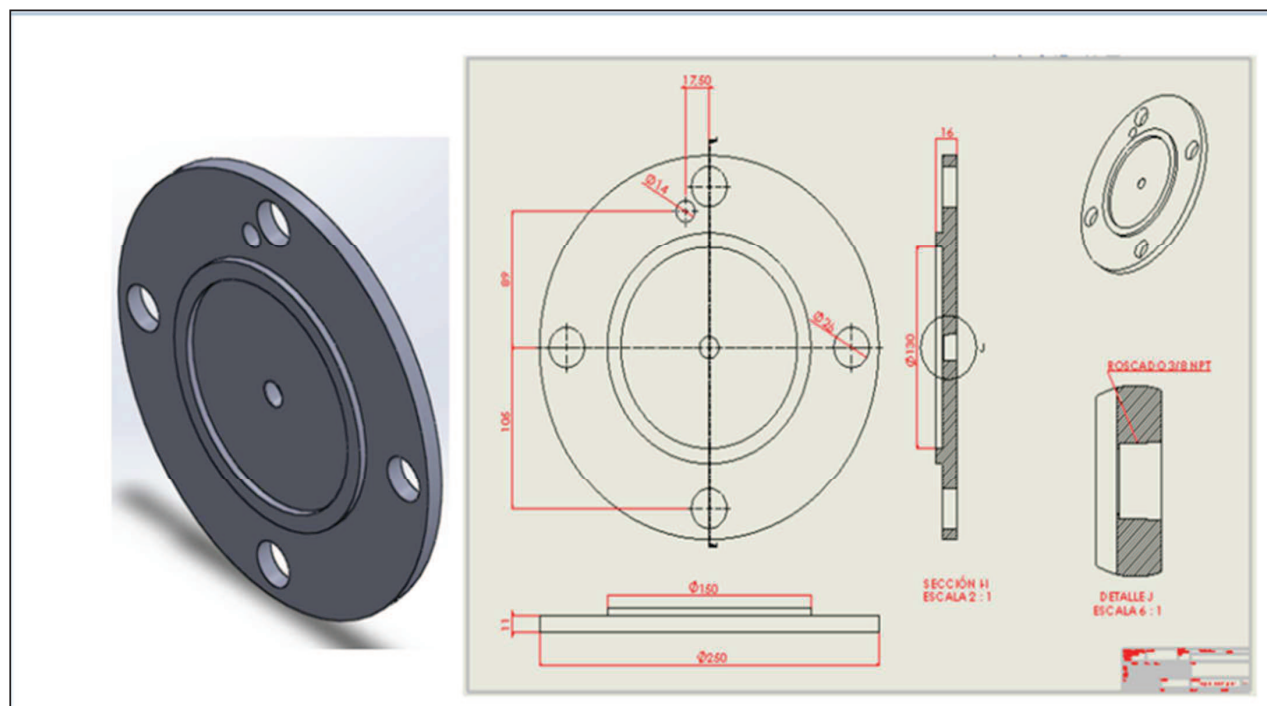
3.1 Acquisition of the items / tools recommended

Follow the list provided in section 2.

3.2 Fabricate the plate adapter

Number 1 in the list of recommended items. Fabricate this plate / adapter according to the following specs.

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3.3 Practical Testing

Before releasing the tools and testing procedure to the normal shop operation, perform the necessary testing to ensure the proper operation and safety of all the steps of the procedure. Make adjustments if necessary.

3.4 Document the SOP

After the testing and final adjustments, document and make available the Standard Operating Procedure (SOP). This SOP must be incorporated into the local library and technical information database of your CRC operation.

3.5 Train the personnel

It is also imperative to train the personnel that will be assigned to test the C-175 dampers. The training must consider the detailed review of the SOP and the practical testing of a damper. Special care to all safety recommendations should be taken.

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3.6 Release the procedure to operation and validate benefits

Release the procedure to normal operation and track the benefits. To determine the final benefits in your operation. Consider determining the % of Redo and the avoidance of affecting the 797 OHT fleet performance results in terms of MTBF, MTBS, MTTR, availability and final cost.

4.0 Benefits

As mentioned before there are direct and indirect benefits associated to this procedure.

Direct Benefits

The CRC operation gets direct benefits from the accurate detection of possible faulty dampers avoiding costly redo work after the C175 Engine is rebuilt and returned to the customers.

Easy detection of faulty dampers when applying the recommended procedure.

Cost effective. The implementation of the recommended procedure requires little investment that, compared with the potential benefits, makes it economically effective.

Indirect Benefits

The indirect benefits are related to avoiding the consequences of installing a faulty C175 damper in an engine that is return to the mine site operation. The damage in availability, MTBF, MTBS, MTTR and maintenance cost that a faulty damper can cause is significant.

The control of unexpected events (failures) in operation at the mine site will contributes to support the production plan and optimize the cost/ton results.

5.0 Resources Required

The resources required are included in sections 2 and 3 of this BP.

6.0 Supporting Attachments / References

The fabrication scheme for the plate adapter (1) is included in section 3 of this BP. If more details are required please contact the authors of this BP.

7.0 Related Best Practices

None

8.0 Acknowledgements

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